

COLD FLARING

Programmed By:- Somnath Vishwakarma

Month- September
2026

PLC INPUT AND OUTPUT DIAGRAM

INPUTS

X0	Start_Watering_unit
X1	Stop_Watering_unit
X2	Emg_Switch_Watering_Unit
X3	Reset_Bit_WT
X4	Head_1_Sensor_WT
X5	Head_3_Sensor_WT
X6	Head_5_Sensor_WT
X7	Head_4_Sensor_WT
X20	Head_2_Sensor_WT
X21	Head_6_Sensor_WT
X22	HEAD_05_BYPASS_WT
X23	HEAD_04_BYPASS_WT
X24	HEAD_03_BYPASS_WT
X25	HEAD_02_BYPASS_WT
X26	HEAD_01_BYPASS_WT
X27	HEAD_06_BYPASS_WT
X30	STOPER_BYPASS_DWT
X31	SCM_Motor_Bypass_PB-02
X32	EMG_DWT
X33	DWT_Clamp_relise_Confirmation
X34	GUN_DOWN_CONFIRMATION
X35	START_DWT
X36	HEAD_04_SENSOR_DWT

INPUTS

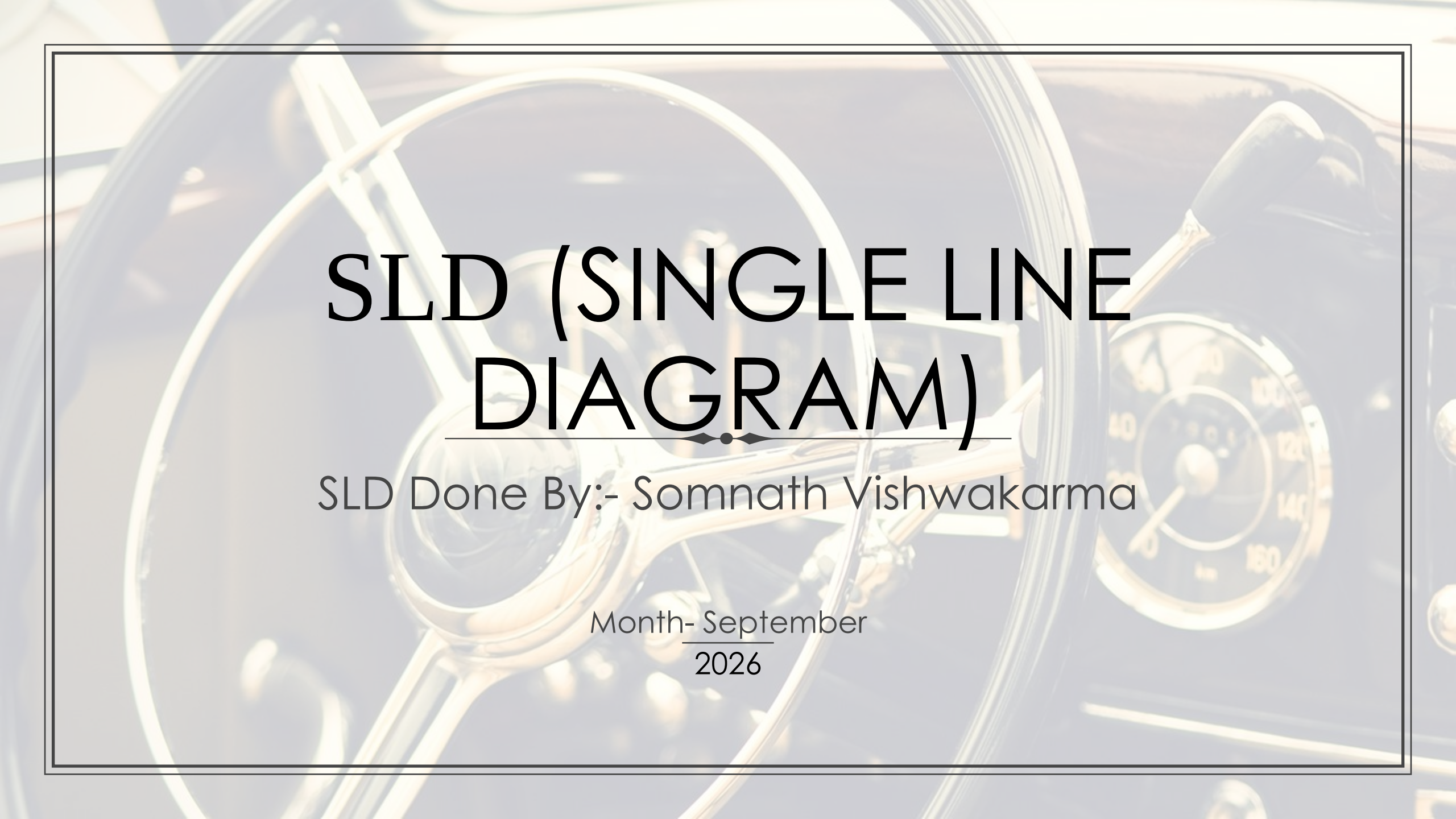
X37	HEAD_03_SENSOR_DWT
X40	HEAD_06_SENSOR_DWT
X41	HEAD_05_SENSOR_DWT
X42	HEAD_01_SENSOR_DWT
X43	HEAD_02_SENSOR_DWT
X44	STOPER_SENSOR_DWT
X45	RST_C4_DWT
X46	PRESSURE_SWITCH_CLD
X47	EMG_CLD
X50	START_CLD
X51	LAST_SENSOR_WT
X52	LAST_SENSOR_CLD
X53	HEAD_05_SENSOR_CLD
X54	HEAD_01_SENSOR_CLD
X55	HEAD_02_SENSOR_CLD
X56	HEAD_04_SENSOR_CLD
X57	HEAD_03_SENSOR_CLD
X60	STOPER_BYPASS_WT
X61	STOPER_BYPASS_CLD
X62	Spare
X63	WATER_MOTOR_RUNNING
X64	STOPER_SENSOR_WT
X65	Suction_Motor_Bypass_On

OUTPUTS

Y0	Head_2_Clamp_With_Lifter_WT
Y1	Head_3_Gun_WT
Y2	Head_2_Gun_WT
Y3	Head_6_Gun_WT
Y4	Head_6_Clamp_With_Lifter_WT
Y5	Head_4_Clamp_With_Lifter_WT
Y20	Head_1_Gun_CLD
Y21	Head_1_Clamp_With_Lifter_WT
Y22	Head_4_Gun_WT
Y23	Head_3_Clamp_With_Lifter_WT
Y24	Stoper_Pre_Watering_unit_WT
Y25	Head_1_Gun_WT
Y26	Head_5_Clamp_With_Lifter_WT
Y27	Head_5_Gun_WT
Y30	Head_3_Clamp_With_Lifter_DWT
Y31	Head_2_Clamp_With_Lifter_DWT
Y32	Head_6_Clamp_With_Lifter_DWT
Y33	Stoper_Pre_Dewatering_unit_DWT
Y34	Head_5_Clamp_With_Lifter_DWT
Y35	Head_1_Clamp_With_Lifter_DWT
Y36	Head_4_Clamp_With_Lifter_DWT
Y37	Green_Light_DWT

OUTPUTS

Y40	Head_4_Gun_CLD
Y41	Head_1_Clamp_With_Lifter_CLD
Y42	Stoper_Pre_Watering_unit_CLD
Y43	Head_3_Gun_CLD
Y44	Head_5_Clamp_With_Lifter_CLD
Y45	Head_4_Clamp_With_Lifter_CLD
Y46	Head_2_Clamp_With_Lifter_CLD
Y47	Head_5_Gun_CLD
Y50	Head_2_Gun_CLD
Y51	Head_3_Clamp_With_Lifter_CLD
Y52	Green_Light_CLD
Y53	RED_LIGHT_CLD
Y54	RED_LIGHT_DWT
Y55	AIR_ROV_DWT
Y56	Filling_Motor_WT_RELAY
Y57	SUCTION_MOTOR_DWT_RELAY
Y60	Gun_Air_ON/OFF_DWT
Y61	END_Stoper_DWT
Y62	ROV_CLD
Y63	Green_Light_WT
Y64	Yellow_Light_for_byypass_condition
Y65	Red_Light_WT

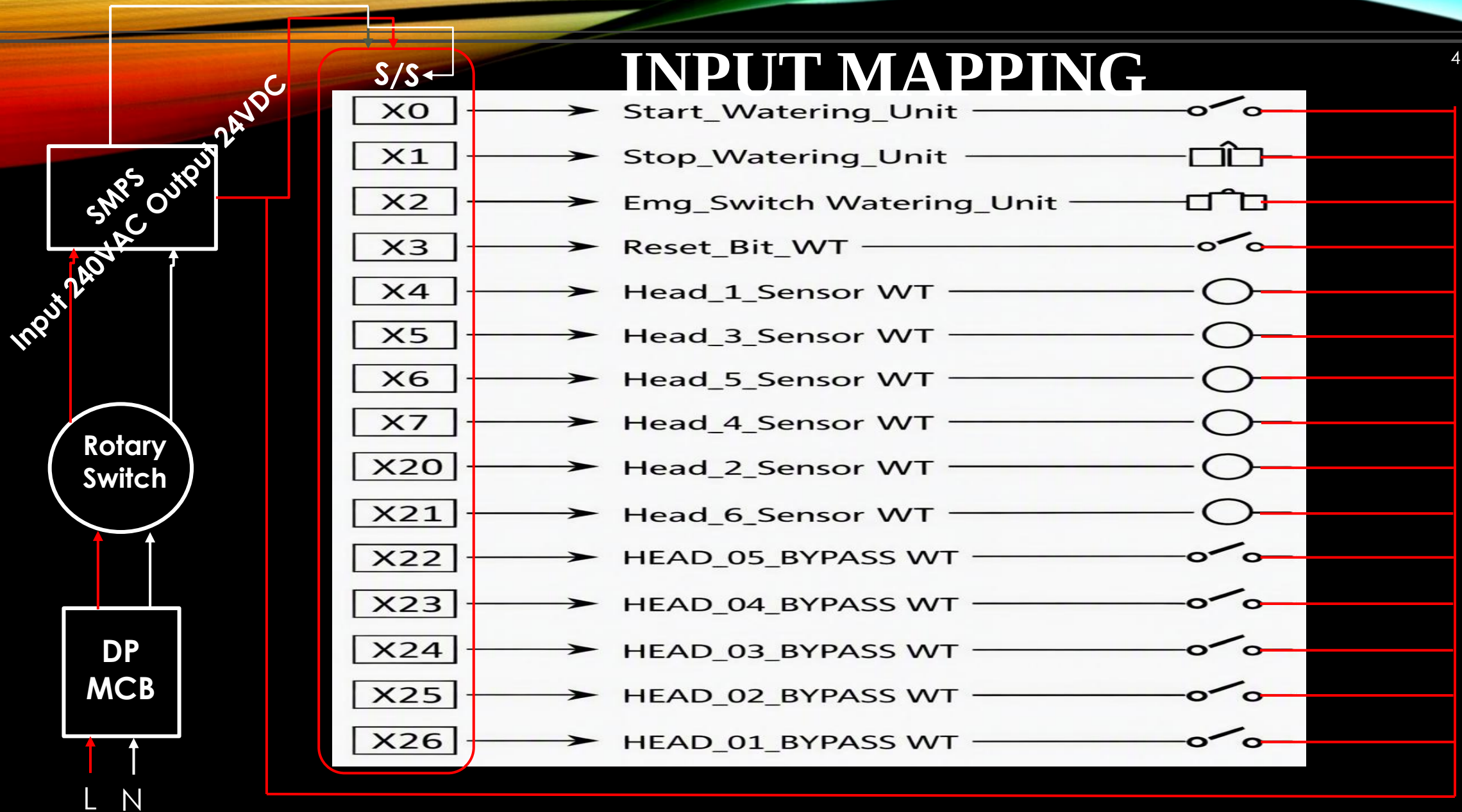


SLD (SINGLE LINE DIAGRAM)

SLD Done By:- Somnath Vishwakarma

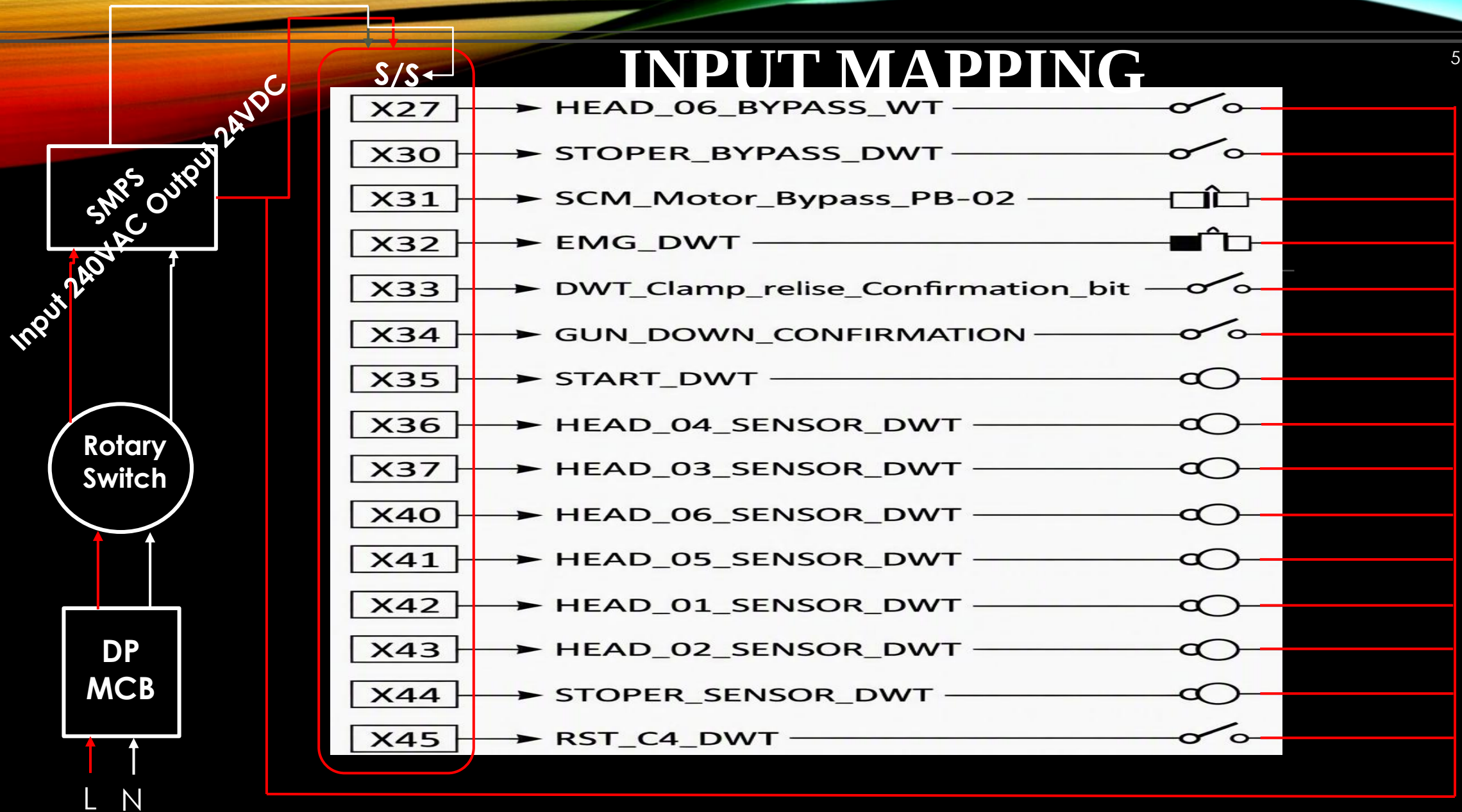
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INPUT MAPPING



INPUT MAPPING

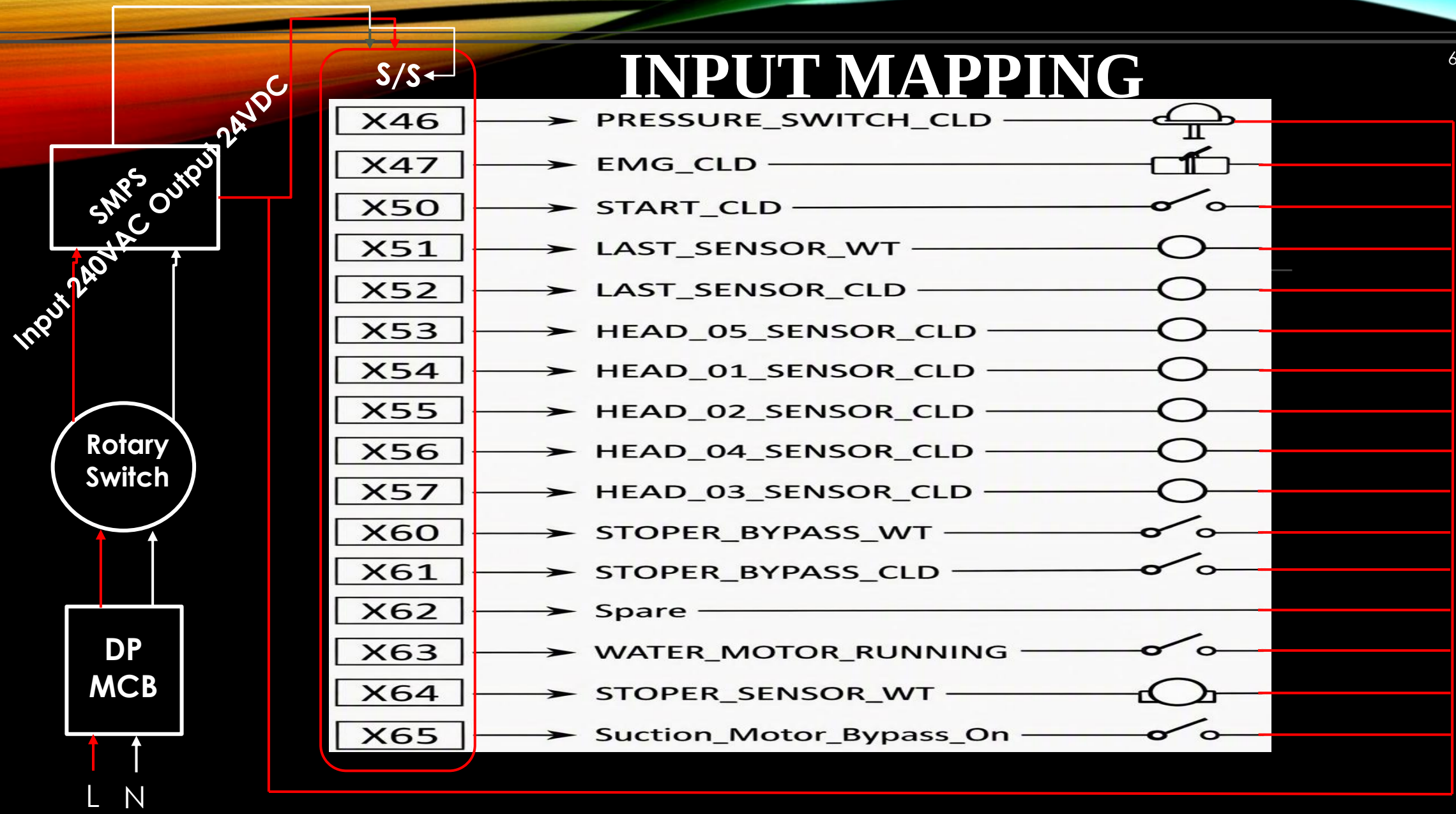
5



INPUT:- X27 TO X45

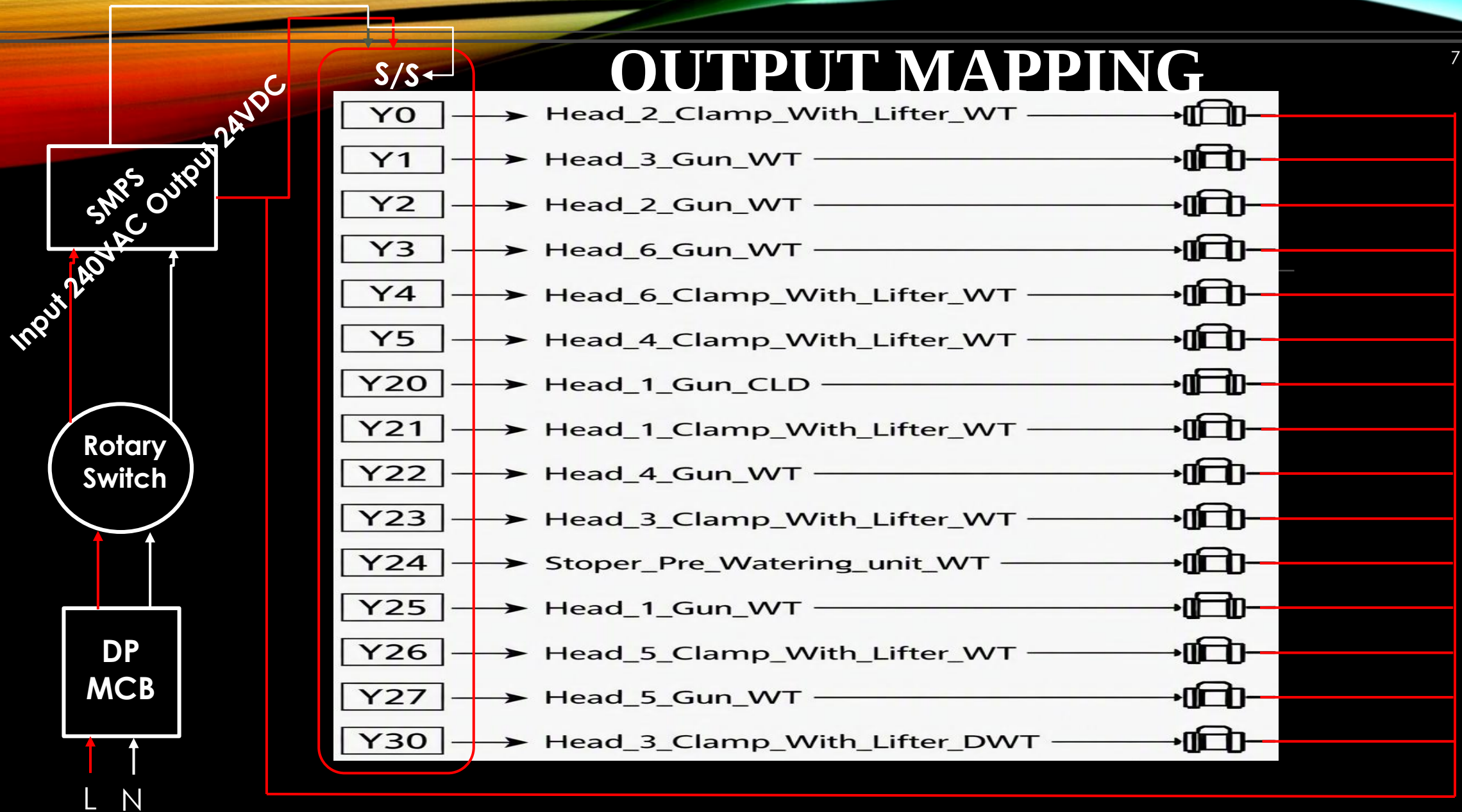
INPUT MAPPING

S/S



OUTPUT MAPPING

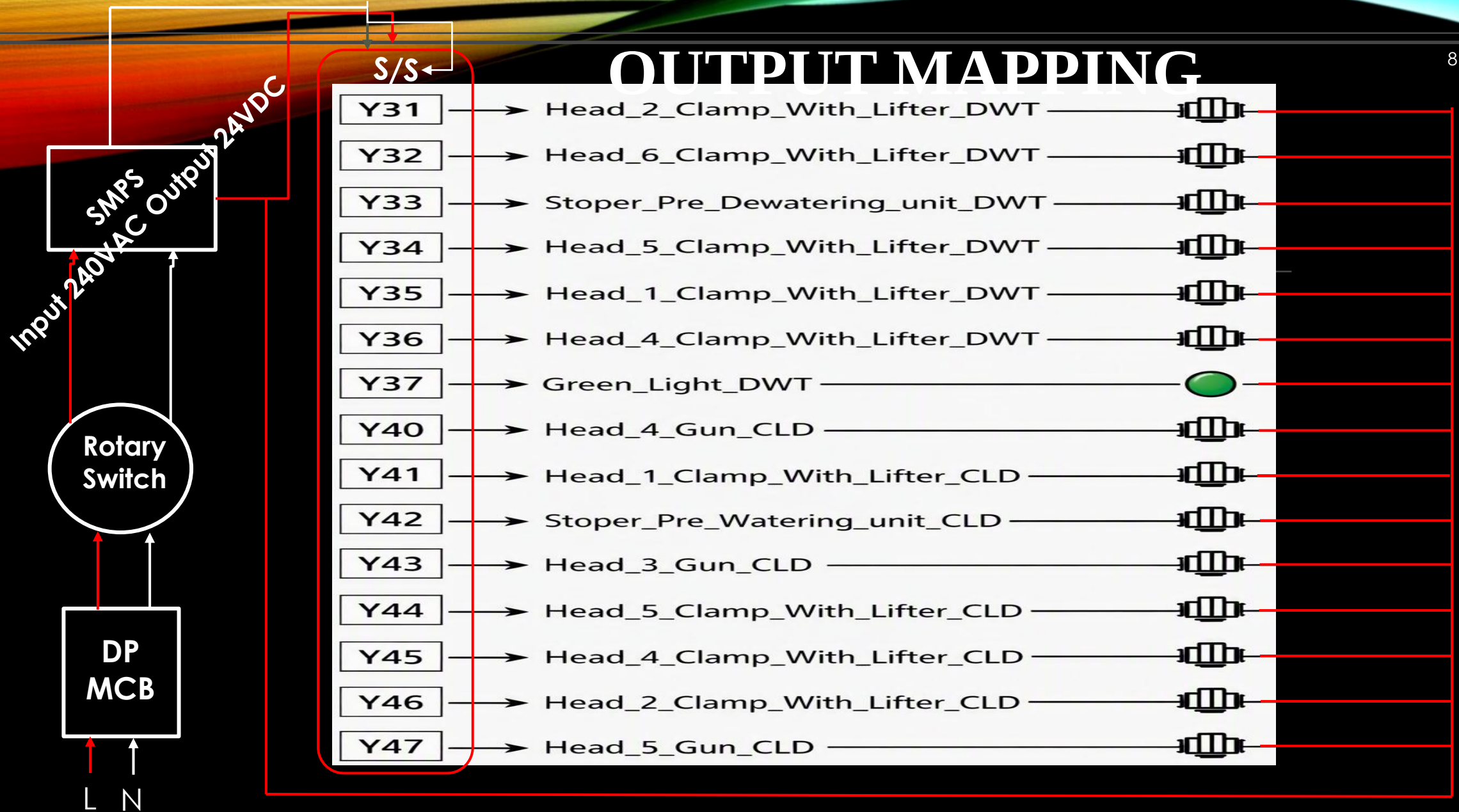
7



OUTPUT:- Y0 TO Y30

OUTPUT MAPPING

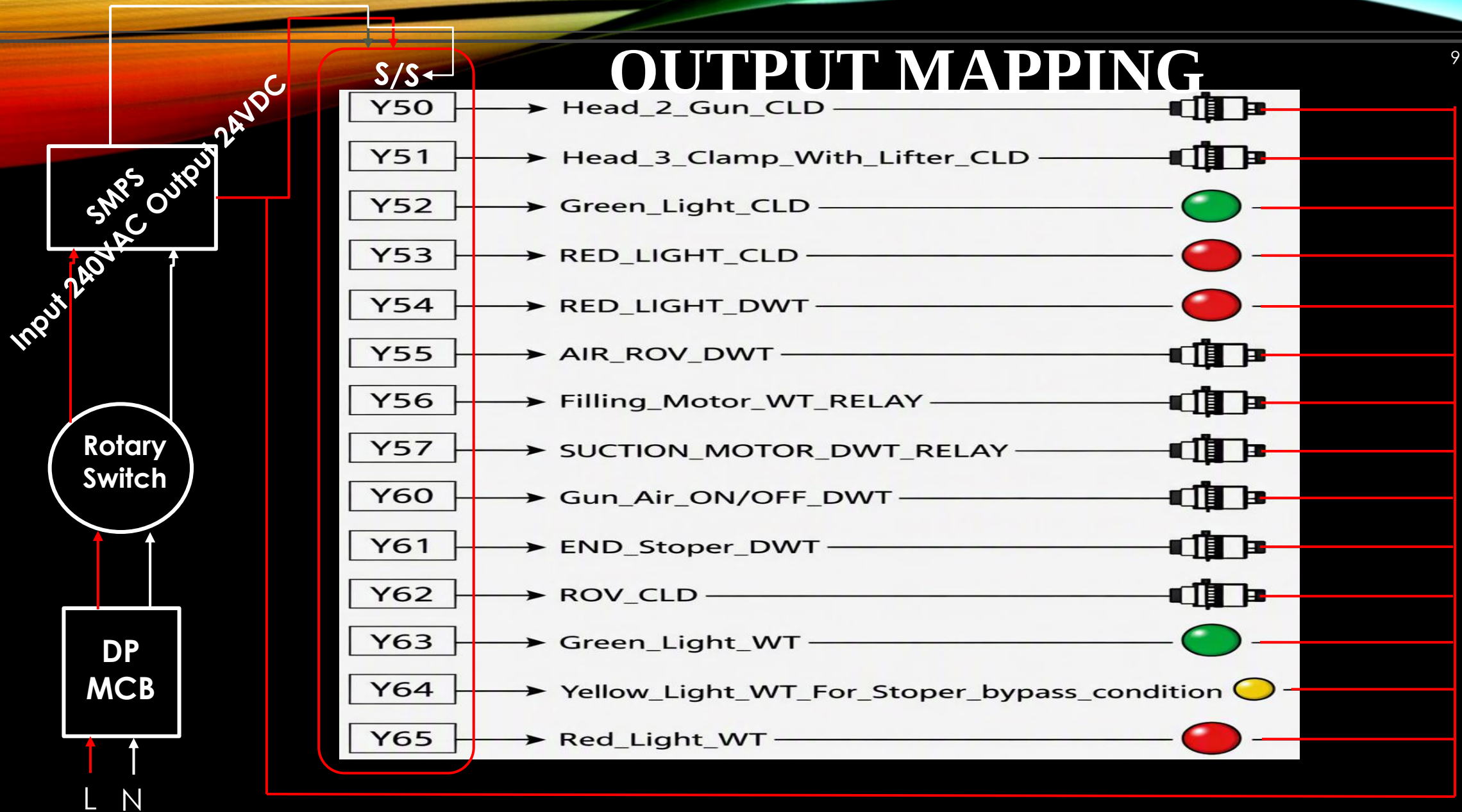
8



OUTPUT:- Y31 TO Y47

OUTPUT MAPPING

9



OUTPUT:- Y50 TO Y65



PANEL DETAILS (MAKE :- SAIEX)

Panel Done By:- Somnath Vishwakarma

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MAKE:- SAIEX



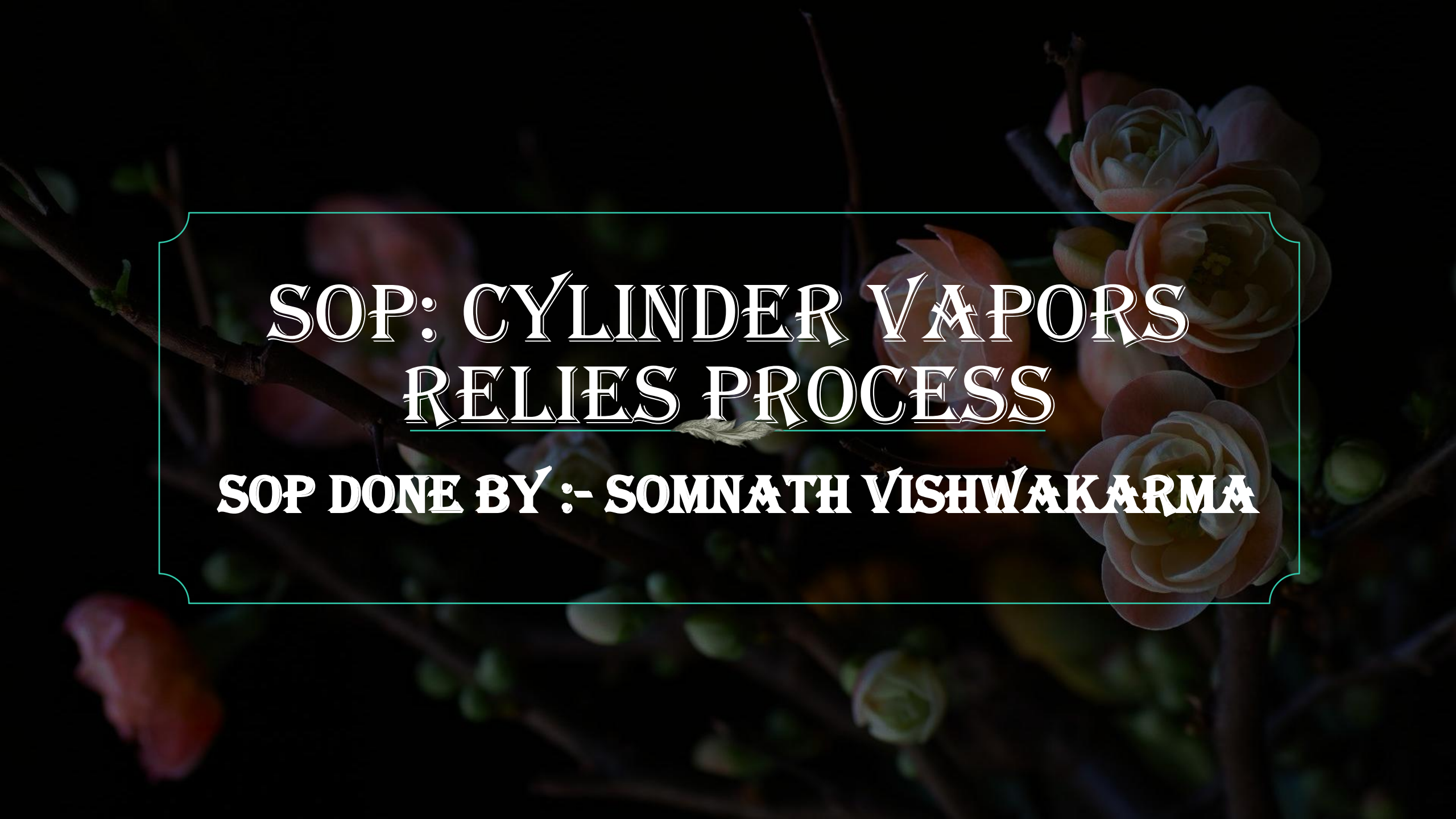


CIMFR DOCUMENT

PESO_1 DOCUMENT

PESO_2 DOCUMENT

Main Panel:- Degassing Shed



SOP: CYLINDER VAPORS RELIES PROCESS

SOP DONE BY :- SOMNATH VISHWAKARMA

COLD FLARING MACHINE START-UP PROCEDURE

01. Check Stoper Bypass Condition

- Before starting the cold flaring machine, verify the stopper bypass condition.
- If the stopper bypass is **ON**, the machine will not start and the green indicator light will remain off.
- Remove the bypass condition, then press **Start**.
- Once cleared, the green LED on the front panel will glow continuously, indicating the machine is ready for automatic operation.

02. Preparation Before Start

- Stop the conveyor.
- Remove all cylinders from inside the machine.
- First start the machine, then start the chain conveyor.

COLD FLARING MACHINE START-UP PROCEDURE

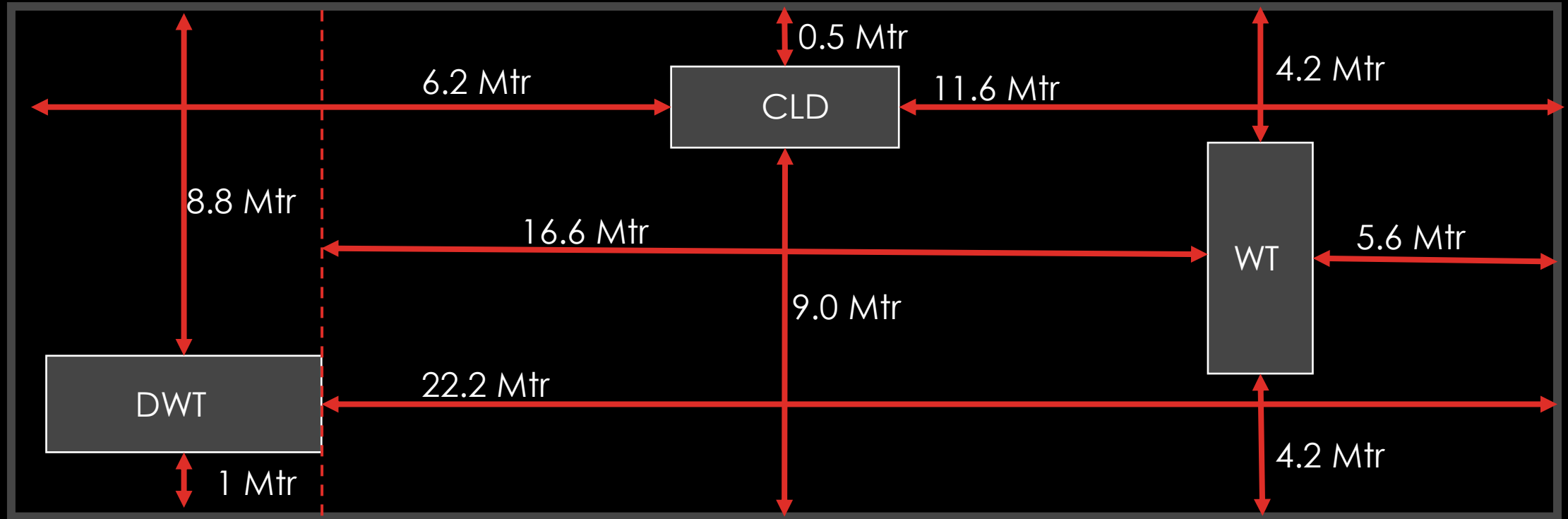
•03. Automatic Operation Sequence

- Cylinders will enter the sensor range one by one.
- The system will automatically clamp the cylinder, lower the gun, and begin flaring.
- If vapor is present inside the cylinder, it will be automatically released.
- Once vapor is cleared, cylinders will be processed one by one in sequence.

•04. Important Notes

- Do not stop the chain conveyor while the machine is running in auto mode.
- If liquid LPG enters the machine, removal may take longer and may not be possible properly.
- In such cases, press the Emergency **(EMG)** Switch, remove all cylinders, and restart the machine following the above procedure.

LAYOUT



Measurement as Lumpsum Data.



THANK YOU

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